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Course – Fundamentals in Artificial Intelligence and Machine Learning

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Domain: Artificial Intelligence (Basic Predictive Systems)
Tools: Python, Matplotlib, Turtle Graphics

1. Abstract

The Daily Water Intake Predictor & Tracker is a simple yet effective application designed to help users maintain proper hydration levels. The system predicts the recommended daily water intake based on individual factors such as body weight, environmental temperature, and activity level.

Unlike generic hydration guidelines, this system adapts recommendations dynamically and provides visual feedback using graphs and graphical interfaces. The integration of prediction logic with visualization tools enhances user awareness and encourages better daily habits.

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2. Problem Statement & Motivation

Maintaining adequate hydration is essential for overall health, yet many individuals fail to monitor their daily water intake.

The major challenges include:

- Lack of awareness regarding required daily water consumption
- Absence of simple tools for tracking intake

- No visual comparison between required and actual consumption
- Difficulty in maintaining consistency in hydration habits

This project aims to solve these issues by providing an interactive and easy-to-use solution.

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3. System Architecture & Methodology

3.1 Prediction Model

The system uses a rule-based predictive model to estimate daily water intake:

- Base calculation depends on body weight
- Adjustments are made based on:
  - Temperature
  - Activity level

This approach ensures that recommendations are personalized and adaptable.

3.2 Data Collection

The user provides:

- Weight (kg)
- Temperature (°C)
- Activity level (low / medium / high)
- Water intake data for the past 5 days

3.3 Visualization Approach

Two types of visualization are implemented:

- **Matplotlib Graph:**
  Displays daily water intake over time and compares it with the recommended level
- **Turtle Graphics Interface:**
  Shows a visual progress bar representing daily completion percentage

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4. Implementation Details

The project is implemented using Python with the following components:

- **Matplotlib:**
  Used to generate line graphs for intake tracking
- **Turtle Graphics:**
  Used to create a visual progress bar

- **Custom Logic:**
  Used for prediction and percentage calculation

The program runs in a terminal-based environment and interacts with the user through inputs.

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5. Results & Discussion

5.1 Output Analysis

The system successfully:

- Predicts daily water intake
- Displays intake trends over multiple days
- Highlights differences between recommended and actual intake

The graphical output provides a clear understanding of hydration patterns.

5.2 Visualization Insights

- The graph helps identify under-consumption or over-consumption
- The turtle progress bar provides an intuitive understanding of daily completion

- Users can easily compare their habits with recommended values

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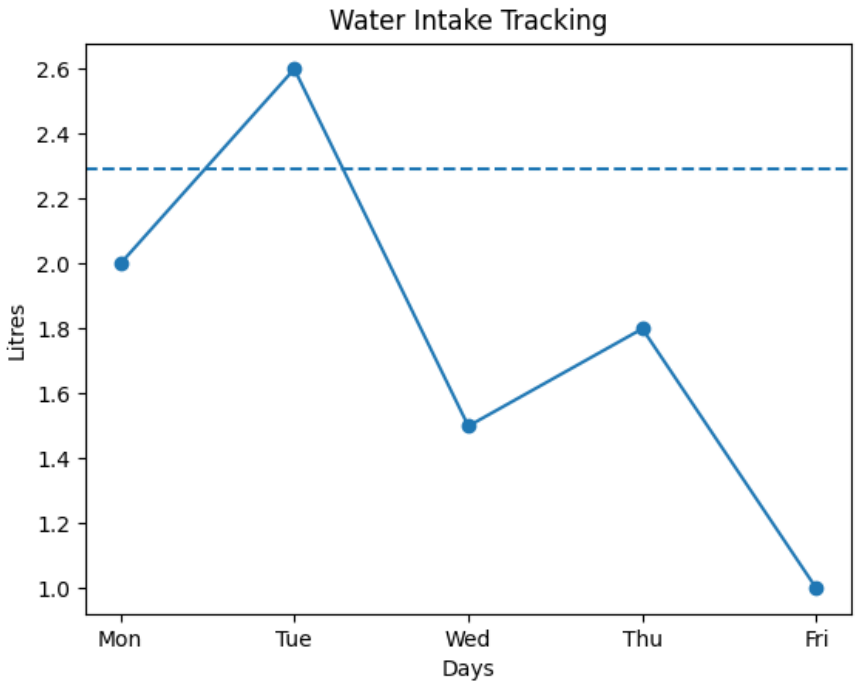
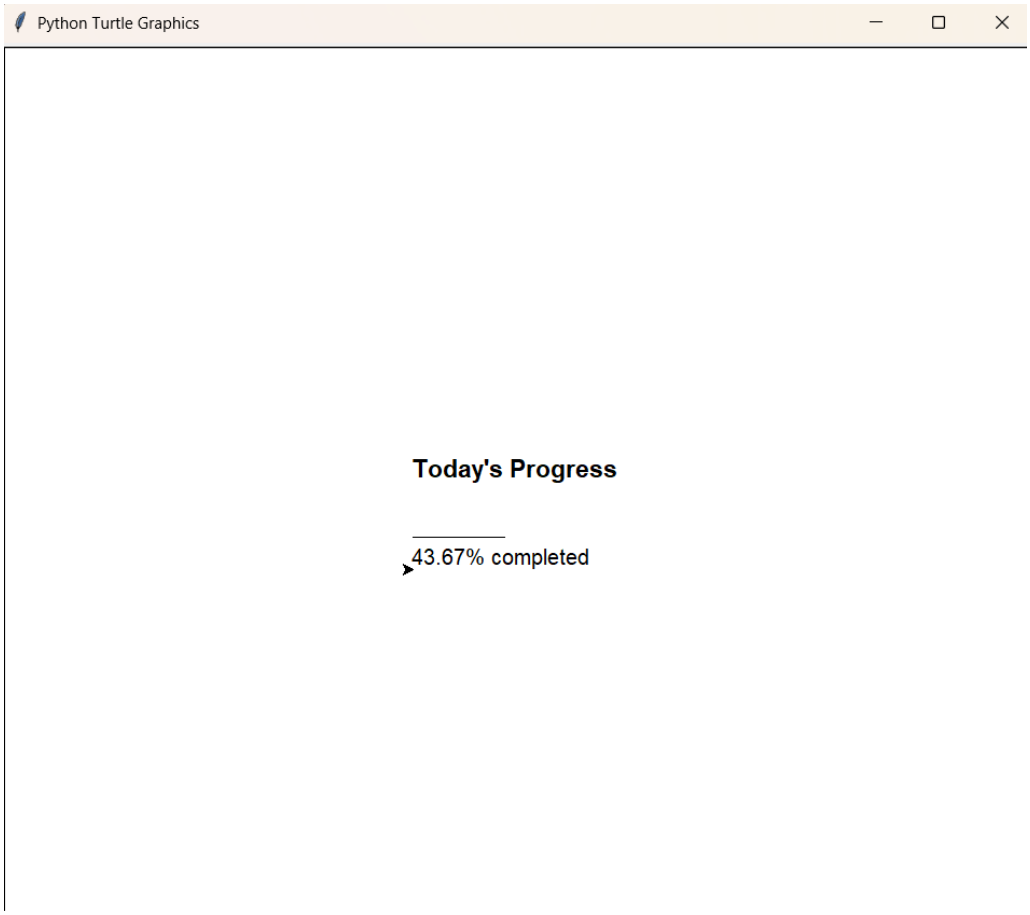
5.3 Output Visualization

📊 Graph Output



Description:
The graph represents water intake over five days and compares it with the recommended intake level. It helps identify patterns and deviations.

🐢 Turtle Visualization



Description:
The turtle interface displays daily progress in percentage form using a visual progress bar, making it easy to understand hydration status.

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6. Challenges Faced

- Integrating multiple libraries (matplotlib and turtle)
- Managing user inputs effectively

- Ensing smooth transition between graph and turtle outputs
- Designing a simple yet meaningful prediction model

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7. Conclusion & Future Scope

This project demonstrates how basic AI concepts can be applied to solve real-life problems. It shows that even simple predictive logic combined with visualization can create useful applications.

Future Enhancements:

- Integration of reminders for hydration
- Development of a graphical user interface (GUI)
- Use of machine learning models for more accurate predictions
- Mobile or web-based application development

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8. Final Summary

The Daily Water Intake Predictor & Tracker successfully bridges the gap between awareness and action in hydration management. By combining

prediction, tracking, and visualization, the system provides a practical solution for improving daily habits.

Final Deliverables

1. Python source code (main.py)
2. Graph visualization output
3. Turtle-based progress visualization
4. GitHub repository with documentation